Jin-Hyeong Park

Machine Learning Engineer at AI Research Institute, Neowiz | Gyeonggi-do, South Korea
Email: jhpark.4745@gmail.com | Google Scholar | Webpage | LinkedIn

RESEARCH INTERESTS

Inverse Rendering, Neural 3D Representation, Neural Relighting

EDUCATION

Chung-Ang University (CAU), Seoul, Korea — Mar 2018 - Feb 2020

M.S. in Computer Science and Engineering *(Focus: Neural Architecture Search)*

- Advisor: Jaesung Lee | GPA 3.82/4.0
- Thesis: Effective Front-end Architecture Search for Random Weight Network on Edge Device

Korea National University of Transportation (KNUT), Gyeonggi-do, South Korea — Mar 2014 - Feb 2018

B.S. in Computer Science and Information Engineering

- Advisor: Sungwook Lee | GPA 3.65/4.0

PUBLICATIONS

Decoupled Reprojection Consistency for Diagnosing 3D Gaussian Splatting Failures
Jin-Hyeong Park
Eurographics 2026 Poster, 2026 [PDF] [Project]

Multi-label Naïve Bayes Classifier Considering Label Dependence
Hae Cheon Kim, ***Jin-Hyeong Park****, Dae Won Kim, Jaesung Lee*
Pattern Recognition Letters, 2020

Compact Feature Subset Based Multi-label Music Categorization for Mobile Devices
Jaesung Lee, Wangduk Seo, ***Jin-Hyeong Park****, Dae Won Kim*
Multimedia Tools and Applications, 2018

PROFESSIONAL EXPERIENCE

AI Research Lab, Neowiz — Nov 2020 - Present

Machine Learning Engineer

- Contributed to the development of the AI portion of an Apple's 2023 Mac Game of the Year Nominated AAA game title (Lies of P)
- Built and deployed diffusion-based image generation workflows and geometry-processing pipelines used across seven game projects.

Automated Machine Learning Lab, **Chung-Ang University (CAU) | PI: Dr. Jaesung Lee**

Graduate Research Assistant — Mar 2018 - Aug 2020

Research Intern — Jul 2017 - Feb 2018

PROJECT EXPERIENCE

Enhanced Game Image Generation AI Service Development and Feature Integration — Jan 2024 - Present

Neowiz, South Korea

Keyword: Image processing, Generative Models, IP-Consistency Generation

- Developed Stable Diffusion-based features for concept art generation, style transfer, and background variation, enhancing the game artwork production workflow.

Automated Hair Guide Model Generation for Game Characters from 3D Reconstruction — Apr 2024 - Nov 2024

Neowiz, South Korea
Keyword: 3D Content Generation, Geometry Processing, Inverse Rendering
- Designed a hair guide model generation pipeline to optimize the creation of game hair models, utilizing Geometry Processing techniques including clustering, importance sampling, and resampling.
- Developed a post-processing procedure that parametrically differentiates guide hair using results from existing Inverse Rendering research.

Development of Facial Animation Pipeline for Lip Sync and Emotion based on Voice and Script Aug 2022 - Oct 2022
Neowiz, South Korea
Keyword: Signal Processing, Multivariate Prediction, Sentiment Analysis
- Designed and implemented an automated pipeline for facial animations in cartoon-style games, utilizing script-based sentiment analysis and speech-to-viseme models to automate facial expression generation

Neural Audio Filter for Transforming Monster Voices into Machinery Sounds Jun 2021 - Oct 2022
Neowiz, South Korea
Keyword: Neural Representation, Signal Processing, GAN-based Style Transfer
- Conducted GAN-based domain transfer to convert monster voices into mechanical sounds; applied in the AAA game Lies of P (Apple's 2023 Mac Game of the Year).

Development of Computer-based Three-Dimensional Medical Image Analysis Program for the Objective Assessment of Orbital Disease Sep 2018 - Oct 2019
National Research Foundation of Korea
Keyword: Computer Vision, **3D CNNs**
- Conducted 3D volumetric data classification using 3D convolutional neural networks, achieving high accuracy in multi-class prediction tasks.

SKILLS

Languages: Python・C++・CUDA・Javascript・LaTeX
Framework: PyTorch・DirectX 11・Flutter
Software: Blender・COLMAP・Unreal Engine・ComfyUI・Git・MATLAB

TEACHING EXPERIENCE

Mentoring for Undergraduate Research Interns, CAU Sep 2018 - Mar 2020
- Mentored experimental design and academic paper composition

Teaching Assistant, CAU
Artificial Intelligence Class Mar 2019 - May 2019
Numerical Analysis Class Sep 2018 - Oct 2018
Discrete Mathematics Class Mar 2018 - May 2018

AWARDS AND HONORS

Funding & Scholarships

Research Assistantship (A), CAU, Full tuition, Mar 2018 - Mar 2020
National Science & Technology Excellence Undergraduate Scholarship, Korea Student Aid Foundation, Full tuition, Sep 2016 - Jun 2017
Academic Excellence Scholarship, KNUT, Sep 2014 - Dec 2014; Mar 2016 - Jun 201

Awards

- **Bronze Medal,** CommonLit Readability Prize, *Kaggle* (Top 10%) 2021
- **Silver Medal,** Ion Switching Prediction, *Kaggle* (Top 4%) 2020